CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 2: Algorithms, Programming and Logic | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 08: PROGRAMMING CONCEPTS

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/22/M/J/24 Q6 (2024 Series)] [Total: 6 marks]

(a) State five standard data types used in programming and give an example value for each. **[2.5 marks]**

(b) Write pseudocode to find the average of 10 numbers stored in a 1D array named Marks[1:10]. **[3.5 marks]**

✓ Marking Scheme / Model Answer:

(a) Integer (e.g. 25), Real (e.g. 19.99), Char (e.g. 'A'), String (e.g. "Computer"), Boolean (TRUE/FALSE) [2.5 marks].
(b) Pseudocode:

```
Total <- 0
FOR i <- 1 TO 10
   Total <- Total + Marks[i]
NEXT i
Average <- Total / 10
OUTPUT "Average is: ", Average [3.5 marks]
```

Question 2 [2210/22/M/J/23 Q4 (2023 Series)] [Total: 6 marks]

(a) A 2D array named StudentScores has 30 rows and 5 columns (StudentScores[1:30, 1:5]). Write pseudocode to find and output the highest score across all students and subjects in this array. **[4 marks]**

(b) State the purpose of using subroutines (functions/procedures) in programming. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) Pseudocode [4 marks]:

```
Highest <- StudentScores[1, 1]
FOR row <- 1 TO 30
   FOR col <- 1 TO 5
      IF StudentScores[row, col] > Highest THEN
         Highest <- StudentScores[row, col]
      ENDIF
   NEXT col
```

NEXT row
OUTPUT "Highest score is: ", Highest

(b) Benefits: Code reusability, modular structure, easier debugging/maintenance, smaller overall program size [2 marks].

Question 3 [2210/21/M/J/24 Q6 (2024 Series)] [Total: 5 marks]

(a) A variable ProductCode has the string value "CS-2024-QP". Explain the output of:

(i) LENGTH(ProductCode)

(ii) SUBSTRING(ProductCode, 4, 4)

(iii) LCASE(SUBSTRING(ProductCode, 1, 2)) **[3 marks]**

(b) State the commands used in pseudocode to open, read from, and close an external text file. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) (i) LENGTH = 10 [1 mark] (ii) SUBSTRING = "2024" [1 mark] (iii) "cs" [1 mark].

(b) OPENFILE "filename.txt" FOR READ; READFILE "filename.txt", Variable; CLOSEFILE "filename.txt" [2 marks].
